

Question

The laboratory uses the following method to analyze the content of a bottle of commercially available resorcinol:

(1) Accurately weigh 1.8928 g of potassium ferricyanide, add 40 mL of water to dissolve, add 2 gKI , 2 mL of hydrochloric acid, let stand for 1 min, then add 10 mL $300 \text{ g}\cdot\text{L}^{-1}$ ZnSO_4 , titrate with $0.2 \text{ mol}\cdot\text{L}^{-1}$ $\text{Na}_2\text{S}_2\text{O}_3$, add 5 mL 0.5% starch after the color becomes lighter, titrate until the blue color disappears, and consume $\text{Na}_2\text{S}_2\text{O}_3$ standard titration solution 26.92 mL

(2) Weigh 0.0966 g of resorcinol, dissolve in 20 mL of water, add 50 mL $\text{pH} = 5$ of HAc-NaAc solution, add 50.00 mL $0.1 \text{ mol}\cdot\text{L}^{-1}$ iodine solution, react for 5 minutes, titrate with the above $\text{Na}_2\text{S}_2\text{O}_3$ standard solution, and consume $\text{Na}_2\text{S}_2\text{O}_3$ standard solution 22.44 mL. At the same time, perform a blank experiment, consuming $\text{Na}_2\text{S}_2\text{O}_3$ standard solution 45.68 mL.

The following statements exist:

1. Phenol can also be titrated using a similar method.
2. Phloroglucinol can also be titrated using a similar method.
3. The sum of the coefficients of the chemical equation in the first step of step (2) is 8.
4. ZnSO_4 is a masking agent to mask the color of $\text{Fe}(\text{CN})_6^{3-}$.
5. The content of resorcinol in this sample is 94.26%.

Which of the above statements are correct (options involving calculations are correct if the error is less than 1%)

A. 1,2

B. 1,3

C. 1,4

D. 1,5

E. 2,3

F. 2,4

G. 2,5

H. 3,4

I. 3,5

J. 4,5

K. 1, 2, 3

L. 1, 2, 4

M. 1, 2, 5

N. 1, 3, 4

O. 1,3,5

P. 1,4,5

Q. 2,3,4

R. 2, 3, 5

S. 2,4,5

T. 3, 4, 5

U. 1, 2, 3, 4

V. 1, 2, 3, 5

W. 1, 2, 4, 5

X. 1,3,4,5

Y. 2, 3, 4, 5

Z. 1, 2, 3, 4, 5

Answer

Correct Answer: **R**

Detailed Explanation

This is a relatively basic iodometric titration question. Among them, step (1) is the titration of $\text{Na}_2\text{S}_2\text{O}_3$ standard solution; step (2) is the titration of the unknown sample of resorcinol.

The analysis is as follows:

For statements 1 and 2, the principle of resorcinol titration in step (2) is that the phenolic hydroxyl group, as an electron-donating group, can undergo aromatic electrophilic substitution reaction, replacing all three H atoms at the ortho- and para- positions with I, thereby performing iodometric back titration.

CHECKPOINT

1 PTS

Phenolic hydroxyl group is an ortho-para directing group and will undergo electrophilic substitution

However, phenol has a low electron-donating ability, making it difficult for aromatic electrophilic substitution to occur at all three positions, so it cannot be used for quantitative analysis; phloroglucinol has a stronger electron-donating ability, so a similar method can be used for titration. Statement 1 is incorrect, statement 2 is correct.

CHECKPOINT

1 PTS

Phenol has a weak electron-donating ability and cannot be used for quantitative analysis; phloroglucinol has a stronger electron-donating ability and can be analyzed

CHECKPOINT

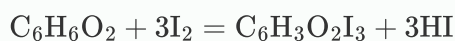
1 PTS

Statement 1 is incorrect, statement 2 is correct

For statement 3, according to the above analysis, the chemical equation that occurs is $\text{C}_6\text{H}_6\text{O}_2 + 3\text{I}_2 = \text{C}_6\text{H}_3\text{O}_2\text{I}_3 + 3\text{HI}$, and the sum of the coefficients of the chemical equation is 8, so statement 3 is correct.

CHECKPOINT

1 PTS



CHECKPOINT

0.5 PTS

The sum of the coefficients of the chemical equation is 8, so statement 3 is correct

For statement 4, ZnSO_4 can react with $\text{Fe}(\text{CN})_6^{4-}$ to form a precipitate, thereby removing the influence of $\text{Fe}(\text{CN})_6^{4-}$ coloration on the iodometric titration endpoint. However, although the statement points out that ZnSO_4 is a masking agent, the masked ion is incorrect, so statement 4 is incorrect.

CHECKPOINT

1 PTS

ZnSO_4 can remove the influence of $\text{Fe}(\text{CN})_6^{4-}$ coloration on the iodometric titration endpoint

CHECKPOINT

0.5 PTS

Masked ion is incorrect, so statement 4 is incorrect

For statement 5, calculate according to the above analysis. The molecular weight of $\text{K}_3\text{Fe}(\text{CN})_6$ is 329.27 g/mol, then $n_{\text{K}_3\text{Fe}(\text{CN})_6} = \frac{1.8928}{329.27} = 0.005748$ mol. According to charge conservation, 1 molecule of $\text{K}_3\text{Fe}(\text{CN})_6$ corresponds to 1 molecule of $\text{Na}_2\text{S}_2\text{O}_3$.

CHECKPOINT

0.5 PTS

1 molecule of $\text{K}_3\text{Fe}(\text{CN})_6$ corresponds to 1 molecule of $\text{Na}_2\text{S}_2\text{O}_3$

Therefore, the amount of $\text{Na}_2\text{S}_2\text{O}_3$ consumed by titration can be calculated as 0.005748 mol, and the concentration of $\text{Na}_2\text{S}_2\text{O}_3$ is $c_{\text{Na}_2\text{S}_2\text{O}_3} = \frac{0.005748}{26.92 \times 10^{-3}} = 0.2135$ mol/L.

From the back titration, it can be calculated that the amount of I_2 consumed in the reaction process is $n_{\text{I}_2} = \frac{1}{2} \times (45.68 - 22.44) \times 10^{-3} \times 0.2135 = 2.481 \times 10^{-3}$ mol. Since 1 molecule of resorcinol corresponds to the consumption of 3 molecules of I_2 , the amount of resorcinol can be calculated as 8.27×10^{-4} mol. The molecular weight of resorcinol is 110.11 g/mol, and the mass of resorcinol can be calculated as 0.09106 g. Thus, the mass fraction of resorcinol is calculated as $\omega = \frac{0.09106}{0.0966} \times 100\% = 94.26\%$, so statement 5 is correct.

CHECKPOINT

2 PTS

Mass fraction of resorcinol is $\omega = 94.26\%$

CHECKPOINT

0.5 PTS

Statement 5 is correct

Therefore, the correct statements are 2, 3, and 5, and option R is correct.

CHECKPOINT

1 PTS

The correct statements are 2, 3, and 5, and option R is correct.